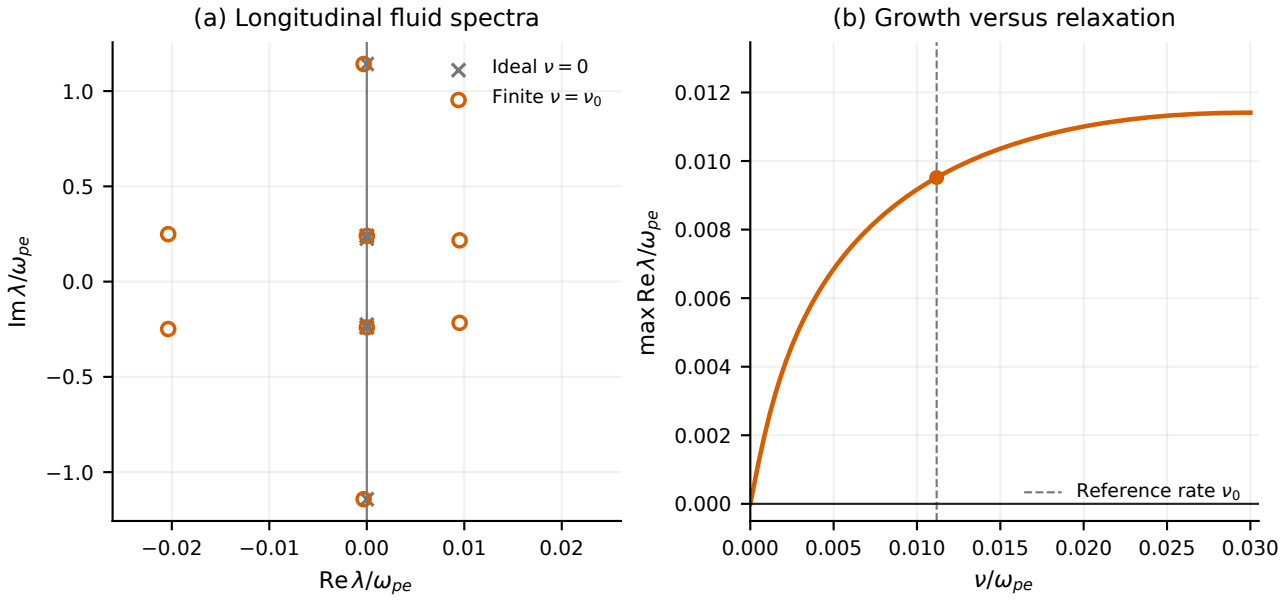
(a) Longitudinal fluid spectra
Ideal $\nu = 0$
Finite $\nu = \nu_0$
$\mathrm{Im}\,\lambda/\omega_{pe}$
$\mathrm{Re}\,\lambda/\omega_{pe}$
1.0
0.5
0.0
−0.5
−1.0
−0.02
−0.01
0.00
0.01
0.02
(b) Growth versus relaxation
$\max \mathrm{Re}\,\lambda/\omega_{pe}$
ν/ω_{pe}
Reference rate ν_0
0.012
0.010
0.008
0.006
0.004
0.002
0.000
0.000
0.005
0.010
0.015
0.020
0.025
0.030